

Group Assignment - Chapter 5, Question 3

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(a) How k -fold cross-validation is implemented

The set of n observations is randomly divided into k groups, or *folds*, of approximately equal size. The procedure then works as follows:

1. One of the k folds is set aside as a validation set, and the statistical method is fit on the remaining $k - 1$ folds.
2. The fitted model is used to predict the observations in the held-out fold, and an error measure (for example, mean squared error MSE_i for regression, or the misclassification rate for classification) is computed on that fold.
3. This process is repeated k times, with a *different* fold serving as the validation set on each repetition, so that every observation is used for validation exactly once and for training $k - 1$ times.

(b) Advantages and disadvantages

(i) Relative to the validation set approach

Advantages of k -fold CV:

- The validation set approach fits the model using only a subset of the observations (the training split), so the resulting test error estimate can be highly variable, depending heavily on exactly which observations happen to fall into the training set versus the validation set.
- Because the validation set approach trains on a smaller subset of the data, it tends to *overestimate* the test error rate that would be obtained if the model were fit on the entire data set, since statistical methods generally perform worse when trained on fewer observations. k -fold CV trains each model on a much larger fraction of the data ($(k-1)/k$ of the observations), so it does not suffer from this bias to the same degree.

Disadvantage of k -fold CV:

- k -fold CV is more computationally expensive, since it requires fitting the model k times rather than just once, as in the validation set approach.

(ii) Relative to LOOCV

Advantages of k -fold CV (e.g. $k = 5$ or $k = 10$):

- LOOCV requires fitting the model n times, once with each single observation left out, which can be very computationally expensive when n is large. k -fold CV with $k < n$ only requires k model fits.
- In LOOCV, the n training sets are all nearly identical to one another. The mean of many highly correlated quantities has higher variance than the mean of quantities that are less correlated. In k -fold CV with $k < n$, the k training sets overlap less with one another, so the resulting estimates are less correlated, and consequently $CV_{(k)}$ has lower variance than the LOOCV estimate.

Disadvantage of k -fold CV:

- **Higher bias.** Each training fold in k -fold CV uses only $(k - 1)/k$ of the observations, a smaller effective sample size than the $n - 1$ observations used in each LOOCV training set. Since models trained on less data tend to have somewhat worse (upwardly biased) test error estimates, k -fold CV incurs more bias than the nearly unbiased LOOCV estimate.

Overall: k -fold CV occupies a middle ground in a bias–variance tradeoff. The validation set approach has both high bias and high variance; LOOCV has low bias but high variance (and is often expensive); k -fold CV with $k = 5$ or $k = 10$ typically achieves a favorable balance — moderate bias and substantially reduced variance compared to LOOCV — which is why it is the most commonly recommended choice in practice.